

The University of Lahore
Department of Computer Science & IT
Mid Term Examination



Spring-2024

Question # 1: (6 marks)

Write down the status Flags (1/0) of the following flags after execution of each piece of code initially all flags are Zero.

INSTRUCTIONS	DESCRIPTION	CF	PF	OF	ZF	SF	AF
OR AH, BL	Where AX=A01Dh, BX=8C18h	0	1	0	0	1	X
NEG BH	Where BX = 7F00h	1	1	0	0	1	1
SAR AX, CL	Where AX = CF23h, CX=8904	0	0	0	0	1	X
ADD AX, DX	Where AX =FFF h, DX = 1h	0	1	0	0	0	1
CMP AL, BH	Where AX=FFF0h, BX = FFFF	1	0	0	0	1	1

Question # 2: (4 marks)

Consider the following Memory Area and give the answers of the following questions:-

0C21:01A0	13 96 D0 E0 D0 E0 A2 1E - 99 80 3E 20 99 00 75 24	> ..u\$
0C21:01B0	A2 24 99 0A C9 75 1D 0A - C0 74 19 8B 0E 21 96 E3	.\$...u...t...!..
0C21:01C0	13 B0 1A 06 33 FF 8E 06 - 00 96 F2 AE 07 75 05 4F	3.....u.O
0C21:01D0	89 3E 21 96 BB 06 97 80 - 3E 13 96 00 74 03 BB 4C	.>!.....>...t..L
0C21:01E0	97 BE C3 96 8B 3E 05 98 - B9 08 00 AC 3C 3F 75 02	>.....<?u.
0C21:01F0	8A 07 3C 20 74 01 AA 43 - E2 F1 B1 03 B0 20 38 04	..< t..C..... 8.
0C21:0200	74 12 B0 2E AA AC 3C 3F-75 02 8A 07 3C 20 74 01	t.....<?u...< t.
0C21:0210	AA 43 E2 F1 32 C0 AA C3-F6 46 04 02 75 43 8B A3	.C..2....F..uC..

- Which Logical address (i.e segment and offset address) in the above Memory area is equal to Physical address 0C3EA? **(01DA)**
- Find the physical address for the 0C21:01FF. **(0C40F)**

Question # 3: Write down assembly language program for the following: (6 marks)

Input: A string which is assumed to be initialized in data segment terminated by "\$".

"5 Quick Brown Fox Jumps Over The 3 Lazy Dogs ☺" (Note : ☺ = "(:)")

Action:

- You don't need to declare other string.
- Read the given string using any addressing mode. Add the numeric values which are being used in the string.
- Perform the case conversion (capital to small and vice versa) for each of the ALPHABET in the above string and replace the original ALPHABET with the converted ALPHABET in the memory.
- Replace the numeric values and ☺ with the space.

Display: Display the modified string and numeric value at the output screen.

```
.MODEL SMALL
.STACK 100H

.DATA
STR DB "5 Quick Brown Fox Jumps Over The 3 Lazy Dogs :)$"
NEWLINE DB 0AH, 0DH, '$'
SUM DB "Sum of the numeric values which are being mentioned in the string = $"
.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX

    ; DISPLAY ORIGINAL STRING
    MOV AH, 09H
    LEA DX, STR
    INT 21H

    ; DISPLAY THE NEWLINE
    MOV DX, OFFSET NEWLINE
    MOV AH, 09H
    INT 21H

    ; READ THE GIVEN STRING AND PERFORM REQUIRED OPERATIONS
    LEA BX, STR ; LOAD THE ADDRESS OF THE STRING
    MOV CX, 0 ; COUNTER FOR STRING LENGTH
    MOV AH, 0 ; CLEAR AH REGISTER
    MOV CX, 0 ; CLEAR CX TO HOLD THE SUM OF NUMERIC VALUES

READ_LOOP:
    MOV AL, [BX] ; LOAD THE CURRENT CHARACTER
    CMP AL, '$' ; CHECK FOR END OF STRING
    JE DONE_READING

    ; CHECK IF THE CHARACTER IS A NUMERIC DIGIT
    CMP AL, '0'
    JB REPLACE_SPECIAL
    CMP AL, '9'
```

JA CHECK_UPPERCASE

**; IF THE CHARACTER IS A DIGIT, ADD ITS VALUE TO BX
SUB AL, '0' ; CONVERT ASCII TO NUMERIC VALUE
ADD CL, AL**

REPLACE_SPECIAL:

**; REPLACE THE DIGIT WITH A SPACE
MOV CH, ' '
MOV [BX], CH
JMP NEXT_CHAR**

CHECK_UPPERCASE:

**CMP AL, 'A'
JB REPLACE_SPECIAL
CMP AL, 'Z'
JBE CONVERSION**

CHECK_LOWERCASE:

**CMP AL, 'a'
JB REPLACE_SPECIAL
CMP AL, 'z'
JBE CONVERSION
JMP REPLACE_SPECIAL**

CONVERSION:

**; CONVERT UPPERCASE TO LOWERCASE AND VICE VERSA
XOR AL, 20H
MOV [BX], AL ; REPLACE THE ORIGINAL CHARACTER WITH THE CONVERTED
CHARACTER**

NEXT_CHAR:

**INC BX ; MOVE TO THE NEXT CHARACTER
JMP READ_LOOP**

DONE_READING:

**; DISPLAY MODIFIED STRING
MOV AH, 09H
LEA DX, STR
INT 21H**

**; DISPLAY SUM OF NUMERIC VALUES
MOV AH, 02H ; FUNCTION TO DISPLAY A CHARACTER
MOV DL, ' ' ; DISPLAY SPACE BEFORE THE SUM
INT 21H**

**; DISPLAY THE NEWLINE
LEA DX, NEWLINE
MOV AH, 09
INT 21H**

**; DISPLAY THE MESSAGE
LEA DX, SUM
MOV AH, 09
INT 21H**

```

; DISPLAY THE SUM
MOV DL, CL ; LOAD SUM INTO DX

ADD DL, '0' ;
MOV AH, 02H ; FUNCTION TO DISPLAY A CHARACTER
INT 21H

; EXIT PROGRAM
MOV AH, 4CH
INT 21H
MAIN ENDP
END MAIN

```

Question # 4: Find the output of the given assembly language (4 Marks)

Consider the Data area of a Program given below, how this data will be stored in the hexadecimal form (address/contents) in Memory? (Assume that VA begins at offset 0010)

VA DB 0ABh, 0CDh, 0EFh, 40h, 50h, 60h

VB DB 123D, 1011B

VC DW 1234h, 0ABCDh

VD DD 45768923h

VE DW 5, 2 Dup("A")

VF DB "CC"

(Note : ASCII code of character C is 43h)

00	...	...	...	...	...	...	...	...	...	...	...	...	...	00	00
10														1E	1F
AB	CD	EF	40	50	60	7B	0B	34	12	CD	AB	23	89	76	45
00	00	...	...	...	...	...	...	...	...	...	...	...	00	00	00
20	21												2D	2E	2F
05	00	41	00	41	00	43	43								
00	00	00	...	...	...	...	...	...	...	...	...	...	...	...	00
30	31	32													3F
00	00	...	...	...	...	...	...	...	...	...	...	...	...	...	00
40	41														4F